OPTIMISED

```
long long solve (vector<int> &A)
    long long mod = 1e9 + 7
    long long n = A.size();
    long long ans = 0;
    for( int b = 0; b < 32; b++) {
        for (int i = 0; i < n; i++) {
            if ( A[i] & (i << b) )one++
        }
        long long zeroes = n - zones;
        ans = (ans + ((ones + zeroes)% Mod) * 2)) % Mod)
}
```

Time Complexity :- O(N*32)

Space Complexity → O(1)

Section :- KRG1-A

Q. We define $f(x, y)$ as number of different bits in ~~integer~~ its binary form. You need to sum of $f(x, y)$ over all pair in a given integer array

Brute force

```
f(x, y):
    ~~while~~
        int num = x ^ y;
        while ( num != 0 ) {
            if ( num & 1 == 1 ) count = (count + 1) % MOD
            num = num >> 1;
        }
    return count
}

main ()

    for( int i = 0; i < N; i++ ) {
        for ( int j = i + 1; j < n; j++ ) {
        ans += f(nums[i], nums[j]) % MOD;
        }
    }
```

Time Complexity → $O(N^2 \log A[i])$

Space Complexity → $O(1)$